

# **Comparative Evaluation of Hybrid Deep Learning Architectures For MI Detection from 12-Lead ECG Signals**

A REPORT SUBMITTED IN  
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INTERNSHIP

By

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## ABSTRACT

This internship work investigates deep learning approaches for binary classification of 12-lead electrocardiogram (ECG) signals into Normal and Myocardial Infarction (MI) classes. The study focuses on three hybrid architectures: CNN–BiLSTM, CNN–BiGRU, and CNN–Positional Encoding–Transformer. A common convolutional feature-extraction stage is used so that the principal architectural difference lies in the subsequent sequence-modelling component. Experiments were conducted using raw ECG signals and signals processed using a primary preprocessing pipeline consisting of band-pass filtering, 50 Hz notch filtering, and Z-score normalization. Additional experiments using the CNN–Positional Encoding–Transformer model examined the effect of individual filtering configurations and modified band-pass cut-off frequencies. The dataset used in the internship experiments contains 10,000 12-lead ECG recordings, with 5,000 Normal and 5,000 MI records. Each recording contains 10 seconds of ECG sampled at 500 Hz. Records containing NaN or missing signal values were excluded during data preparation. The experiments show that preprocessing does not necessarily improve classification performance for every architecture. Among the reported raw-signal experiments, CNN–BiLSTM achieved the highest observed accuracy of 95.54%, while the preprocessed CNN–BiLSTM experiment achieved 94.25% accuracy. The results support the importance of evaluating preprocessing and sequence-modelling choices jointly rather than assuming that denoising always improves downstream classification.

**Keywords:** ECG, Myocardial Infarction, Deep Learning, CNN, BiLSTM, BiGRU, Transformer, Positional Encoding, ECG Preprocessing, Denoising

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## LIST OF ABBREVIATIONS

| Abbreviation | Meaning                                                  |
|--------------|----------------------------------------------------------|
| ECG          | Electrocardiogram                                        |
| MI           | Myocardial Infarction                                    |
| CNN          | Convolutional Neural Network                             |
| BiLSTM       | Bidirectional Long Short-Term Memory                     |
| BiGRU        | Bidirectional Gated Recurrent Unit                       |
| ROC-AUC      | Receiver Operating Characteristic – Area Under the Curve |
| MCC          | Matthews Correlation Coefficient                         |
| RMSE         | Root Mean Square Error                                   |
| PRD          | Percentage Root-Mean-Square Difference                   |
| SNR          | Signal-to-Noise Ratio                                    |

# Chapter 1: Introduction

## 1.1 Background

Electrocardiography (ECG) is a non-invasive technique for recording the electrical activity of the heart. ECG waveforms contain morphological and temporal information that can support the identification of cardiac abnormalities. Myocardial Infarction (MI) is associated with inadequate or interrupted blood supply to the myocardium, and the resulting electrical changes may be reflected in ECG morphology. For this reason, ECG is an important signal source for automated MI detection.

MI-related changes may appear in the ST segment, Q/R waves, and T waves, and their distribution across leads can provide information about the affected myocardial region. Because a 12-lead ECG captures electrical activity from multiple viewpoints, multi-lead analysis is useful for distinguishing normal signals from MI-associated patterns.

## 1.2 Problem Statement

Automated Normal-versus-MI classification from ECG signals must handle both local morphological characteristics and temporal dependencies. In addition, ECG recordings may contain baseline drift, power-line interference, motion-related components, and other unwanted signal variations. A preprocessing operation can improve signal quality, but excessive or inappropriate filtering may also modify clinically relevant waveform information. Therefore, there is a need to evaluate both model architecture and preprocessing configuration in a controlled manner.

The problem addressed in this internship is the comparative evaluation of CNN-based hybrid architectures for binary classification of 12-lead ECG signals into Normal and MI classes, together with an investigation of the effect of raw versus preprocessed inputs and different filtering configurations, and band-pass cut-off frequencies.

## 1.3 Aim of the Internship Project

The aim of the internship project is to develop and experimentally compare CNN-based hybrid deep learning architectures for binary classification of Normal and MI 12-lead ECG signals, while studying how ECG preprocessing and filtering configurations affect classification performance.

## 1.4 Objectives

- Prepare a balanced set of Normal and MI 12-lead ECG records for binary classification.
- Implement a common CNN-based feature extraction stage for the three main models.
- Develop CNN–BiLSTM, CNN–BiGRU, and CNN–Positional Encoding–Transformer architectures.
- Compare raw ECG input with the primary preprocessed ECG input.
- Evaluate additional filtering and band-pass cut-off configurations using the Transformer-based model.
- Analyze training behaviour and classification performance using standard evaluation metrics.

## 1.5 Scope of the Work

The work is limited to binary Normal-versus-MI classification using 12-lead ECG recordings. The primary experiments compare raw and preprocessed inputs under otherwise comparable model settings. Additional filtering experiments are used to examine the sensitivity of the Transformer-based model to preprocessing choices. The work is experimental and does not attempt to replace clinical diagnosis.



## 1.6 Organization of the Report

Chapter 1 introduces the problem and objectives. Chapter 2 summarizes the ECG and deep learning background relevant to the implementation. Chapter 3 describes the proposed methodology and system design. Chapter 4 presents dataset preparation, preprocessing, model implementation, and experimental setup. Chapter 5 presents the results and discussion. Chapter 6 concludes the work and identifies possible future extensions.

## Chapter 2: Literature Review and Background

### 2.1 Domain Overview

ECG classification combines signal processing with pattern recognition. Traditional approaches often use manually extracted features such as R-peak, QRS-complex, ST-segment, T-wave, and wavelet-based characteristics followed by conventional machine learning. Deep learning reduces dependence on manual feature design by learning representations directly from the signal.

### 2.2 ECG Denoising and Signal Preprocessing

ECG recordings are susceptible to baseline wander, power-line interference, electrode motion, muscle activity, and other artifacts. Previous work has investigated filtering, transform-based techniques, decomposition methods, convolutional denoising autoencoders, recurrent denoisers, and generative approaches. Chiang et al. demonstrated a fully convolutional denoising autoencoder for ECG noise reduction[4], while later work explored recurrent and autoencoder-based approaches for more realistic wearable and reconstruction settings[3],[6]. These studies motivate the use of controlled preprocessing experiments rather than assuming that a filtered signal is always superior for downstream classification.

### 2.3 Deep Learning for MI Detection

Deep convolutional models have been used to detect MI directly from ECG signals without requiring extensive manual feature extraction. Acharya et al. reported an 11-layer CNN for normal-versus-MI ECG beat classification[8], and Liu et al. investigated a 13-layer CNN using three-second Lead II segments[9]. These studies show the ability of CNNs to learn discriminative ECG representations, while also highlighting the role of signal quality and experimental configuration.

### 2.4 Hybrid and Transformer-Based Architectures

Hybrid architectures combine local feature extraction with sequence modelling. CNN layers can learn local waveform patterns, whereas recurrent networks such as BiLSTM and BiGRU can model temporal relationships. Transformer architectures use self-attention to represent relationships among sequence elements, with positional encoding providing information about sequence position. Recent ECG research has increasingly examined attention-based and multimodal Transformer models[11], although many such studies address multi-label cardiovascular prediction or incorporate structured clinical information beyond the ECG waveform.

### 2.5 Research Gap / Motivation

The reviewed literature demonstrates strong progress in both ECG denoising and automated disease classification. However, reconstruction-oriented denoising metrics do not directly indicate whether preprocessing improves the final classification task. In addition, comparisons between different sequence-modelling architectures can be confounded when their feature-extraction stages differ. This internship therefore uses a CNN-based feature-

extraction followed by three alternative sequence-modelling components under raw and preprocessed input conditions.

## Chapter 3: Problem Analysis and Proposed Methodology

### 3.1 System Overview

The system receives a 12-lead ECG record represented as a multivariate temporal signal. The input can be supplied either in its raw form or after the primary preprocessing pipeline. The signal is passed through CNN feature extraction, followed by one of three sequence-modelling components: BiLSTM, BiGRU, or positional encoding with a Transformer encoder. A classification head produces a binary Normal/MI prediction.

### 3.2 Requirement Analysis

- ECG records readable in the selected WFDB-compatible data-processing workflow.
- Numerical processing tools for signal manipulation and normalization.
- Filtering routines for band-pass and notch filtering.
- A deep learning framework for model development and training.
- Visualization tools for training curves and ECG comparisons.
- Consistent input dimensions, class labels, and train/validation/test separation.
- Reproducible evaluation using common classification metrics.

### 3.3 Proposed Methodology

The methodology consists of five stages. First, the selected ECG records are organized according to the Normal and MI labels. Second, records with invalid or missing signal values are removed during data preparation. Third, raw ECG signals are retained for the baseline experiments while the primary preprocessing pipeline is applied for the preprocessed condition. Fourth, the CNN-based feature extraction is followed by three sequence-modelling alternatives. Finally, the trained models are evaluated using accuracy, precision, recall/sensitivity, specificity, F1-score, ROC-AUC, and MCC where available.

### 3.4 Workflow / Framework

ECG recording → data preparation → raw or optional preprocessing → CNN feature extraction → sequence modelling → classification head → Normal/MI prediction → metric-based evaluation.

### 3.5 Tools and Technologies Used

- Python for data processing and model implementation.
- WFDB-compatible ECG data handling for ECG records.
- NumPy and related numerical tools for signal processing.
- SciPy-style digital filtering operations for band-pass and notch filtering.
- TensorFlow/Keras-style deep learning components for CNN, recurrent, and Transformer layers.
- Matplotlib for ECG and training-curve visualization.

## 3.6 Algorithm / Model Description

### 3.6.1 CNN-BiLSTM

The CNN-BiLSTM model used in the experiments combines convolutional feature extraction with bidirectional sequence modelling for binary Normal/MI classification. The overall architecture of the CNN-BiLSTM model is illustrated in Figure 3.1. The model uses three convolutional blocks with 32, 64, and 128 filters and kernel sizes of 7, 5, and 3, respectively. Batch normalization, MaxPooling1D and dropout are incorporated within the convolutional feature-extraction stage.

The resulting feature sequence is then processed using a 64-unit bidirectional LSTM with `return_sequences=True`, allowing information from both forward and backward temporal directions to be considered. The sequence representation is subsequently passed through Global Average Pooling, followed by a dense layer with 128 ReLU units and a dropout layer. Finally, a one-unit sigmoid output layer produces the binary Normal/MI classification.

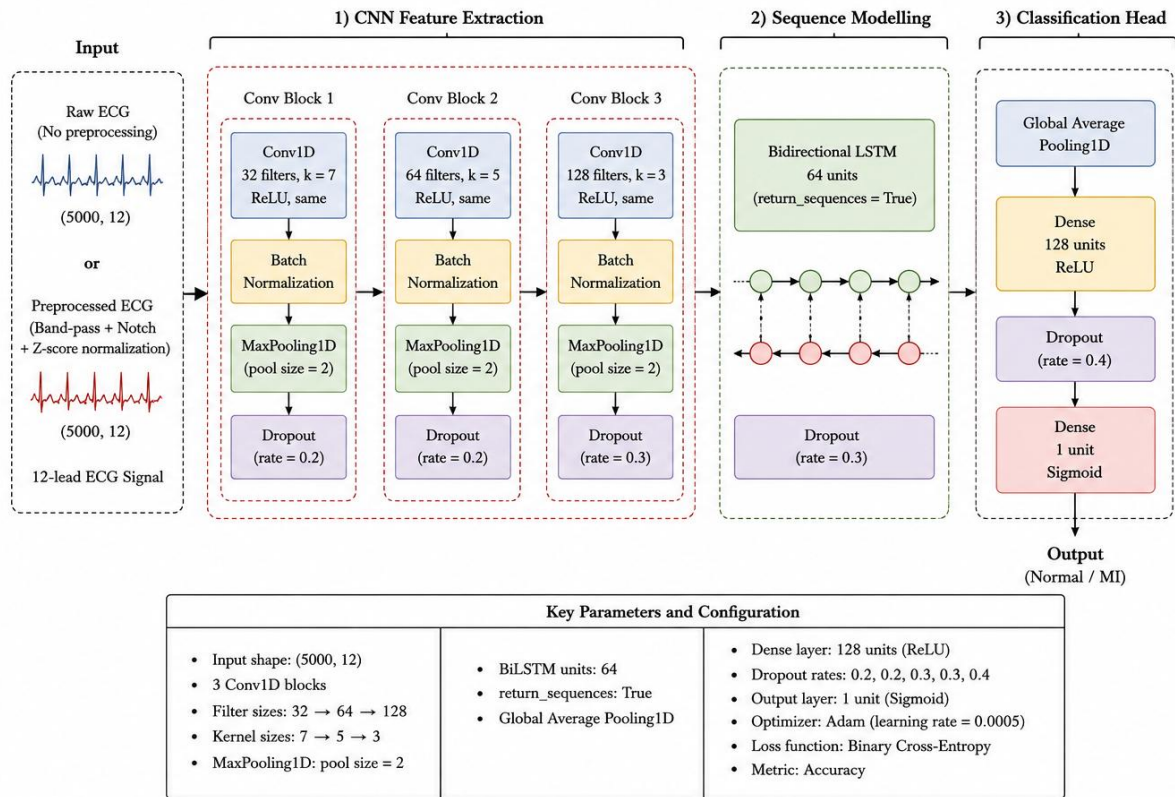


Figure 3.1. CNN-BiLSTM architecture used in the experiments.

Figure 3.1 shows the complete flow of the ECG signal from the input stage through CNN-based feature extraction, BiLSTM sequence modelling and the final classification head. The architecture supports both raw ECG input and the selected preprocessed ECG input consisting of band-pass filtering, notch filtering and Z-score normalization.

### 3.6.2 CNN–BiGRU

The CNN–BiGRU model combines convolutional feature extraction with bidirectional GRU-based sequence modelling for binary Normal/MI classification. The overall architecture of the CNN–BiGRU model is illustrated in Figure 3.2. The model uses two convolutional blocks with 64 and 128 filters and kernel sizes of 7 and 5, respectively. Each convolutional block consists of Conv1D, batch normalization, MaxPooling1D and dropout layers.

The extracted feature sequence is then passed to a 64-unit bidirectional GRU with `return_sequences=False`. A dropout layer is applied after the BiGRU layer to reduce overfitting. The resulting representation is passed directly to a one-unit sigmoid classification layer, which produces the binary Normal/MI prediction.

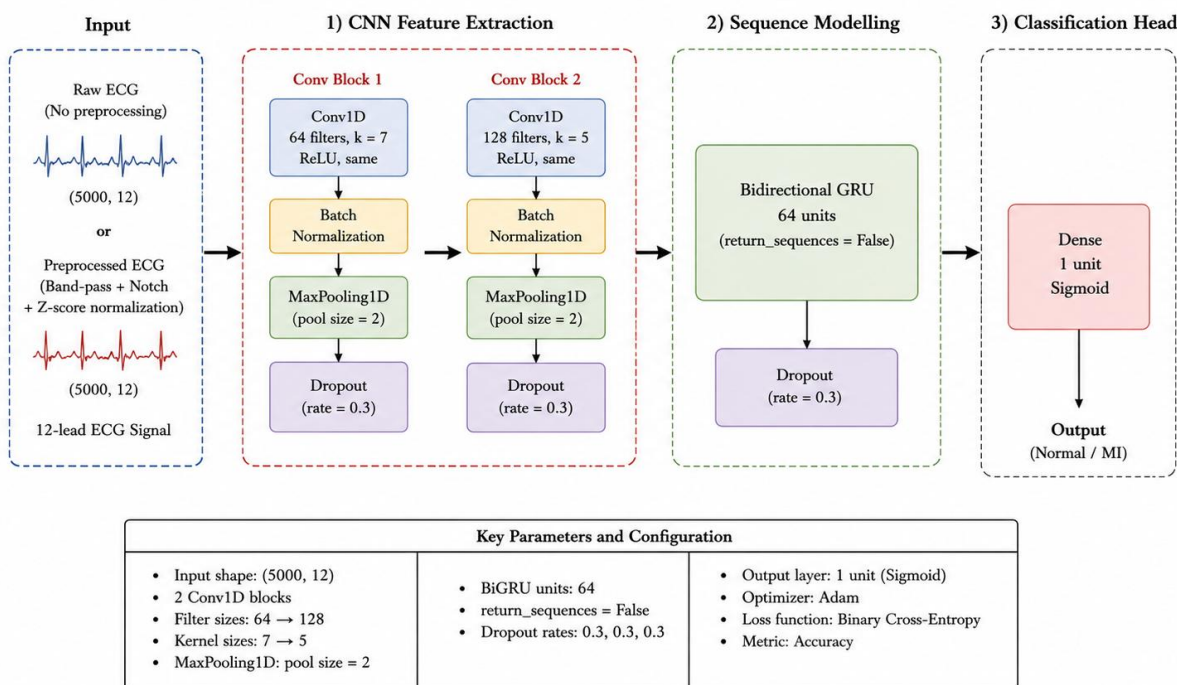


Figure 3.2. CNN–BiGRU architecture used in the experiments.

Figure 3.2 shows the complete flow of the ECG signal through the CNN-based feature-extraction blocks, bidirectional GRU sequence modelling and the final classification head. The architecture accepts both raw ECG signals and the selected preprocessed ECG signals consisting of band-pass filtering, notch filtering and Z-score normalization.

### 3.6.3 CNN–Positional Encoding–Transformer

The CNN with Positional Encoding–Transformer model combines convolutional feature extraction with positional encoding and Transformer-based sequence modelling for binary Normal/MI classification. The overall architecture of the CNN with Positional Encoding–Transformer model is illustrated in Figure 3.3. The model uses two convolutional blocks with 64 and 128 filters and kernel sizes of 7 and 5, respectively. Each convolutional block incorporates batch normalization, MaxPooling1D and dropout for feature extraction and regularization.

The resulting feature sequence is projected into a 64-dimensional representation and combined with sinusoidal positional encoding to retain positional information within the sequence. The encoded representation is then processed using a Transformer encoder consisting of a multi-head self-attention module with four attention heads and a feed-forward network. Dropout and layer normalization are incorporated within the Transformer blocks.

The output of the Transformer encoder is passed through Global Average Pooling, followed by a dense layer with 64 ReLU units and dropout. Finally, a one-unit sigmoid output layer produces the binary Normal/MI classification.

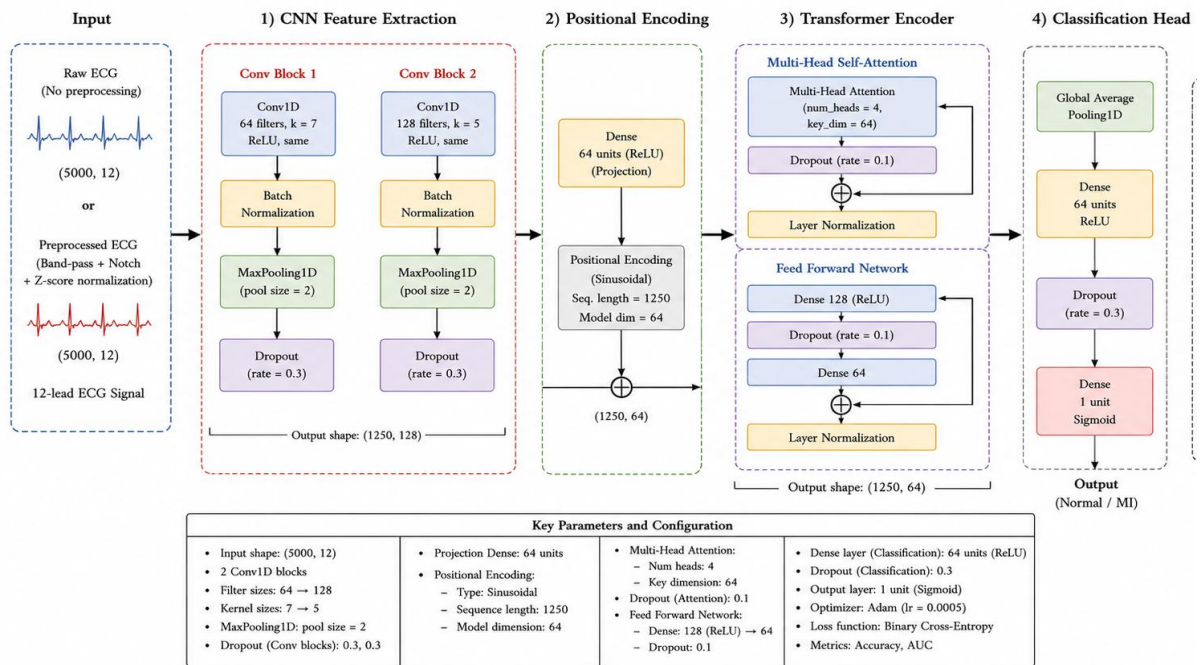


Figure 3.3 CNN-Positional Encoding-Transformer architecture used in the experiments.

Figure 3.3 shows the complete flow of the ECG signal through CNN-based feature extraction, positional encoding, Transformer-based sequence modelling and the final classification head. The architecture supports both raw ECG input and the selected preprocessed ECG input consisting of band-pass filtering, notch filtering and Z-score normalization.



## Chapter 4: Implementation

### 4.1 Dataset / Data Collection

The ECG records used in this internship study were obtained from the MIMIC-IV 1.0 dataset[13]. The dataset was used for ECG signal analysis and subsequent deep learning based classification experiments. A total of 10,000 12-lead ECG records were selected for binary classification, consisting of 5,000 Normal and 5,000 Myocardial Infarction (MI) records. Each recording contains 10 seconds of ECG sampled at 500 Hz, corresponding to 5,000 samples per lead across 12 leads. Thus, each model input is represented with an effective dimension of  $5,000 \times 12$ .

During data preparation, records containing NaN values or missing signal values were excluded to avoid invalid inputs during model training. The remaining records were organized according to their class labels and used consistently for the raw and preprocessed experiments.

### 4.2 Data Pre-processing

Preprocessing was applied independently to each ECG lead. For the primary preprocessed condition, each raw signal was first passed through a 0.5–40 Hz band-pass filter, followed by a 50 Hz notch filter. The filtered signal was then normalized using Z-score normalization. The resulting preprocessed signal was supplied to the deep learning models. The raw condition was retained separately so that the effect of preprocessing could be evaluated without changing the model architecture.

For Z-score normalization, the normalized value is calculated as  $z = (x - \mu) / \sigma$ , where  $\mu$  is the mean and  $\sigma$  is the standard deviation of the corresponding lead. If  $\sigma$  is zero, it is replaced by one to avoid division by zero.

### 4.3 Model Development / System Design

All three main models were developed around a common CNN feature-extraction principle. This controlled design was used to make the comparison focus mainly on the sequence-modelling component. The BiLSTM, BiGRU, and Transformer components were connected to classification heads producing a binary Normal/MI output.

### 4.4 Implementation Details

The main implementation components and their corresponding configurations are summarized in Table 4.1

Table 4.1. Main Implementation components and configuration

| Component    | Configuration                                         | Purpose                            |
|--------------|-------------------------------------------------------|------------------------------------|
| Input        | 5,000 samples $\times$ 12 leads                       | 12-lead ECG representation         |
| CNN blocks   | Conv1D + Batch Normalization + MaxPooling1D + Dropout | Local feature extraction           |
| BiLSTM       | 64 bidirectional units                                | Temporal sequence modelling        |
| BiGRU        | 64 bidirectional units                                | Temporal sequence modelling        |
| Transformer  | 64-dimensional projection, 4 attention heads          | Attention-based sequence modelling |
| Output       | Dense + Sigmoid, 1 unit                               | Binary Normal/MI classification    |
| Optimization | Adam; binary cross-entropy                            | Model training                     |

## 4.5 Experimental Setup

The main experiments were conducted under two input conditions: raw ECG and preprocessed ECG. The model architecture and evaluation procedure were kept unchanged within each corresponding comparison. Additional experiments were performed with the CNN–Positional Encoding–Transformer model using individual filtering configurations and modified band-pass cut-off frequencies. Training and validation accuracy/loss curves were recorded for each experiment to inspect learning behaviour and possible generalization gaps.

## Chapter 5: Results and Discussion

### 5.1 Performance Evaluation Metrics

The experiments use standard binary classification metrics. Accuracy measures the proportion of correctly classified samples. Precision measures the proportion of predicted positive samples that are actually positive. Recall or sensitivity measures the proportion of MI samples correctly detected. Specificity measures the proportion of Normal samples correctly identified. F1-score is the harmonic mean of precision and recall. ROC-AUC summarizes discrimination across classification thresholds, while MCC provides a balanced correlation-based measure for binary classification.

### 5.2 Main Experimental Results

The experimental results show that model performance varies with the input condition and architecture. Among the evaluated configurations, CNN-BiLSTM with raw ECG input achieved the highest overall performance, with 95.54% accuracy, 95.02% recall, 95.53% F1-score, and 0.9893 ROC-AUC. The corresponding denoised condition achieved 94.25% accuracy, while CNN-BiGRU and CNN-Positional Encoding-Transformer also showed comparable performance under raw and denoised conditions. The detailed performance comparison of the evaluated models under raw and denoised ECG conditions is summarized in Table 5.1.

#### 5.2.1. Model Performance Comparison

Table 5.1. Performance comparison of the three CNN-based architectures under raw and denoised ECG input conditions.

| Model / Input                                     | Accuracy | Precision | Recall | Specificity | F1 Score | ROC AUC | MCC    |
|---------------------------------------------------|----------|-----------|--------|-------------|----------|---------|--------|
| CNN + BiLSTM (Raw ECG)                            | 95.54%   | 96.05%    | 95.02% | 96.06%      | 95.53%   | 0.9893  | 0.9108 |
| CNN + BiLSTM (Denoised)                           | 94.25%   | 95.95%    | 92.46% | 96.06%      | 94.17%   | 0.9854  | 0.8857 |
| CNN + BiGRU (Raw ECG)                             | 94.05%   | 96.32%    | 91.66% | 96.47%      | 93.93%   | 0.9848  | 0.8821 |
| CNN + BiGRU (Denoised/Preprocessed)               | 93.17%   | 94.46%    | 91.79% | 94.57%      | 93.11%   | 0.9806  | 0.8638 |
| CNN + Positional Encoding + Transformer (Raw ECG) | 93.85%   | 95.15%    | 92.46% | 95.24%      | 93.79%   | 0.9867  | 0.8773 |

|                                                    |        |        |        |        |        |        |        |
|----------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|
| CNN + Positional Encoding + Transformer (Denoised) | 93.78% | 94.53% | 93.00% | 94.57% | 93.76% | 0.9811 | 0.8757 |
|----------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|

### 5.2.2. Filtering / Cut-off Experiments

Additional experiments were conducted with CNN + Positional Encoding + Transformer to study the effect of filter selection and cut-off values. Table 5.2 summarizes the classification performance obtained with different filtering configurations and cut-off frequency settings.

Table 5.2. Performance comparison of filtering/ cut-off experiments

| <b>Filtering configuration</b>  | <b>Accuracy</b> | <b>Precision</b> | <b>Recall / Sensitivity</b> | <b>Specificity</b> | <b>F1 Score</b> | <b>ROC AUC</b> | <b>MCC</b> |
|---------------------------------|-----------------|------------------|-----------------------------|--------------------|-----------------|----------------|------------|
| Band-pass only (0.5–40 Hz)      | 92.02%          | 95.36%           | 88.43%                      | 95.65%             | 91.76%          | 0.9766         | 0.8427     |
| Notch only                      | 92.29%          | 96.18%           | 88.16%                      | 96.47%             | 91.99%          | 0.9804         | 0.8489     |
| Low-cut 0.67 Hz, High-cut 40 Hz | 93.44%          | 95.88%           | 90.85%                      | 96.06%             | 93.30%          | 0.9847         | 0.8701     |
| Low-cut 0.5 Hz, High-cut 45 Hz  | 93.85%          | 95.28%           | 92.33%                      | 95.38%             | 93.78%          | 0.9819         | 0.8774     |

The filtering experiments show that the choice of cut-off frequencies affects the classification performance. Among the tested configurations, the 0.5-45 Hz band-pass setting achieved the highest accuracy(93.85%), F1-score(93.78%), and MCC(0.8774), while the 0.67-40 HZ configuration achieved the highest ROC-AUC (0.9847). The results indicate that the selected filtering range can influence the classification performance of the Transformer-based model.



### 5.3 Training and Validation Curves

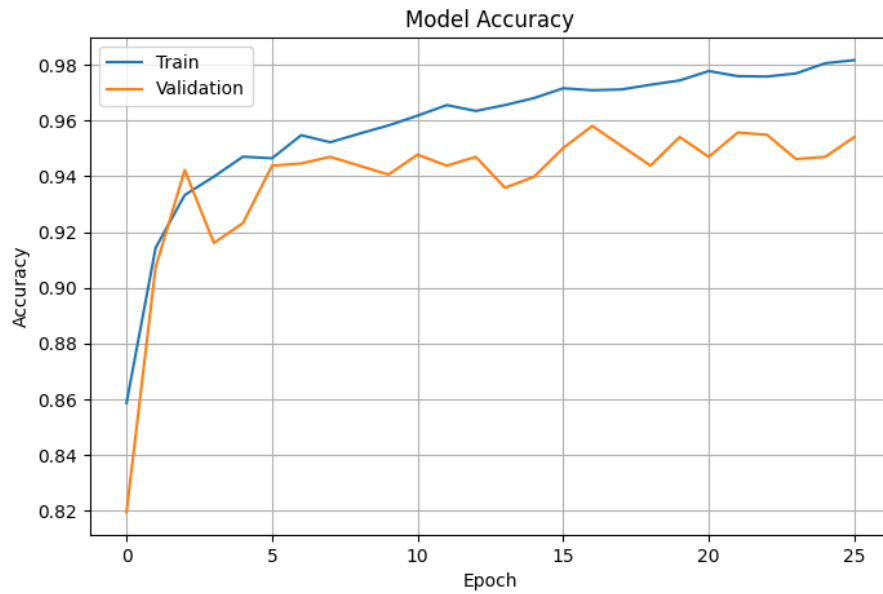


Figure 5.1. Training and validation accuracy of CNN-BiLSTM using raw ECG signals.

Figure 5.1 shows the training and validation accuracy of the CNN-BiLSTM model using raw ECG signals. The training accuracy increases progressively and reaches approximately 98%, while the validation accuracy remains around 94–96% during the later epochs. The relatively small difference between the curves indicates good learning and generalization performance.

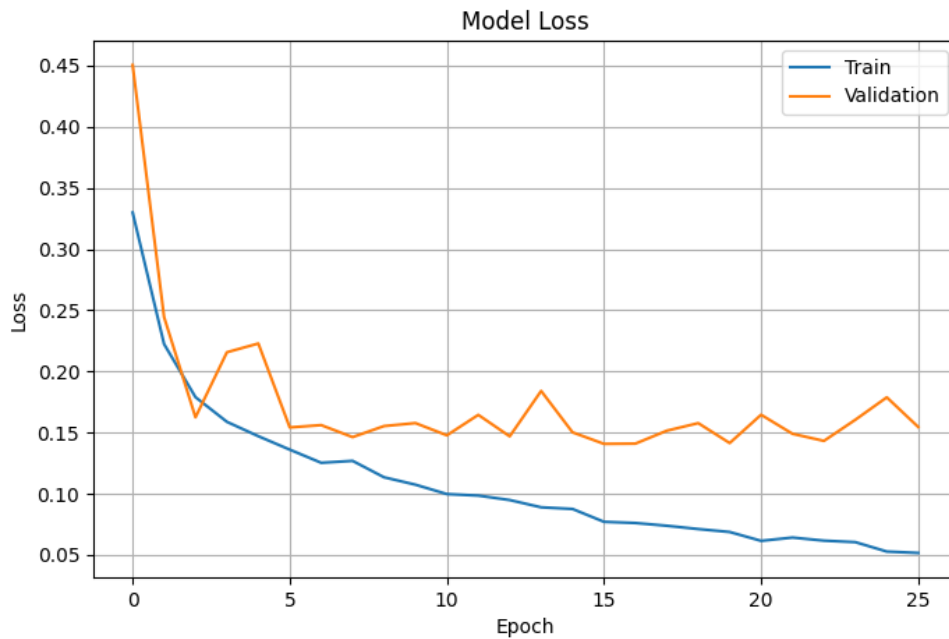
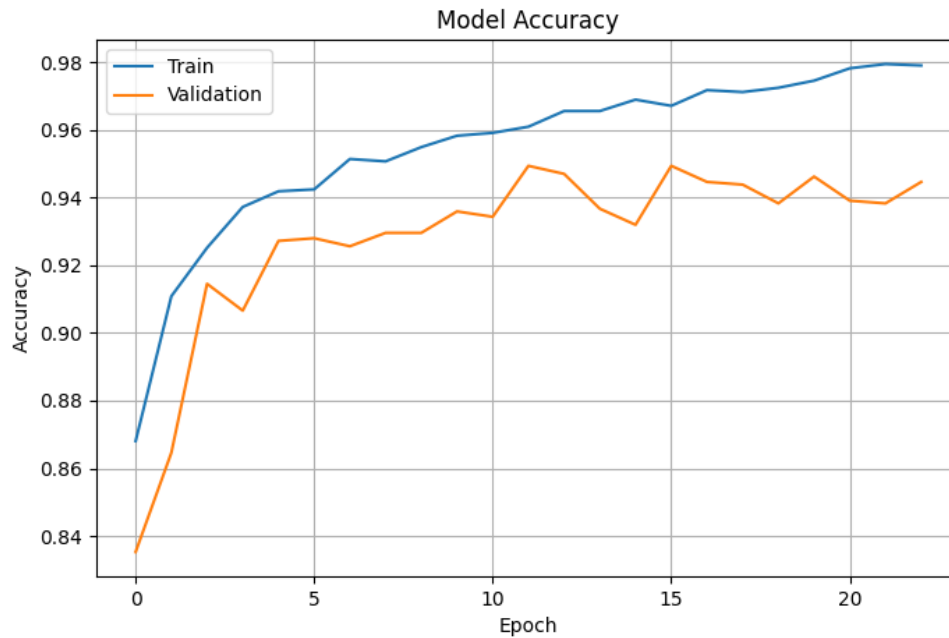


Figure 5.2. Training and validation loss of CNN-BiLSTM using raw ECG signals.

Figure 5.2 shows the training and validation loss of the CNN–BiLSTM model using raw ECG signals. The training loss decreases steadily throughout the training process, while the validation loss decreases initially and then shows moderate fluctuations. This indicates effective learning of the model, with some variation in validation performance during the later epochs.



*Figure 5.3. Training and validation accuracy of CNN–BiLSTM using preprocessed ECG signals.*

Figure 5.3 shows the training and validation accuracy of the CNN–BiLSTM model using preprocessed ECG signals. The training accuracy increases steadily and reaches approximately 98%, while the validation accuracy remains around 93–95% during the later epochs. The curves indicate that the model learns effectively from the preprocessed ECG signals, with a moderate gap between training and validation accuracy.

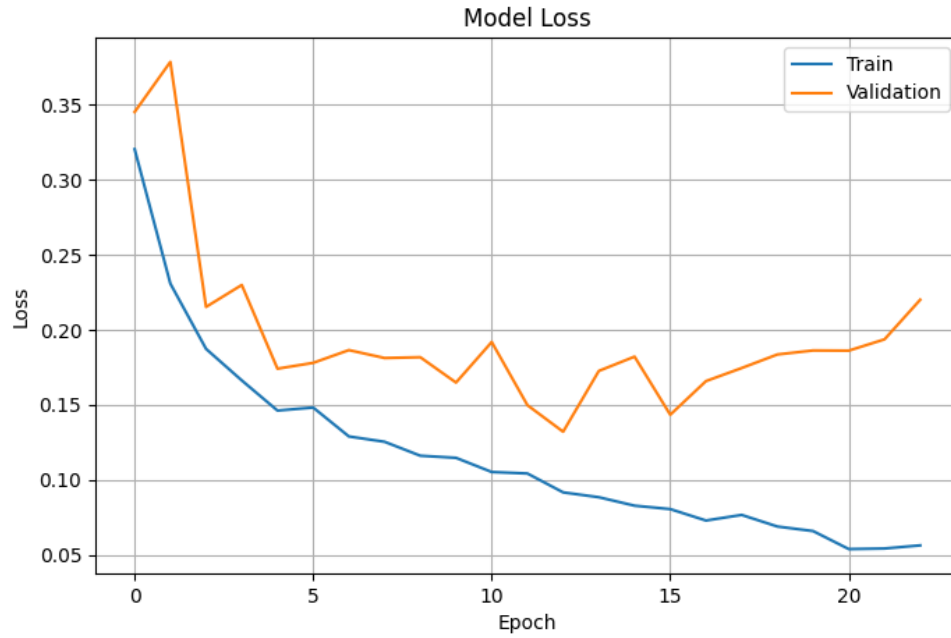


Figure 5.4. Training and validation loss of CNN–BiLSTM using preprocessed ECG signals.

Figure 5.4 shows the training and validation loss of the CNN–BiLSTM model using preprocessed ECG signals. The training loss decreases consistently throughout the training process, whereas the validation loss decreases initially and then fluctuates, with a slight increase towards the final epochs. This indicates continued improvement in training performance with some variation in validation behaviour.

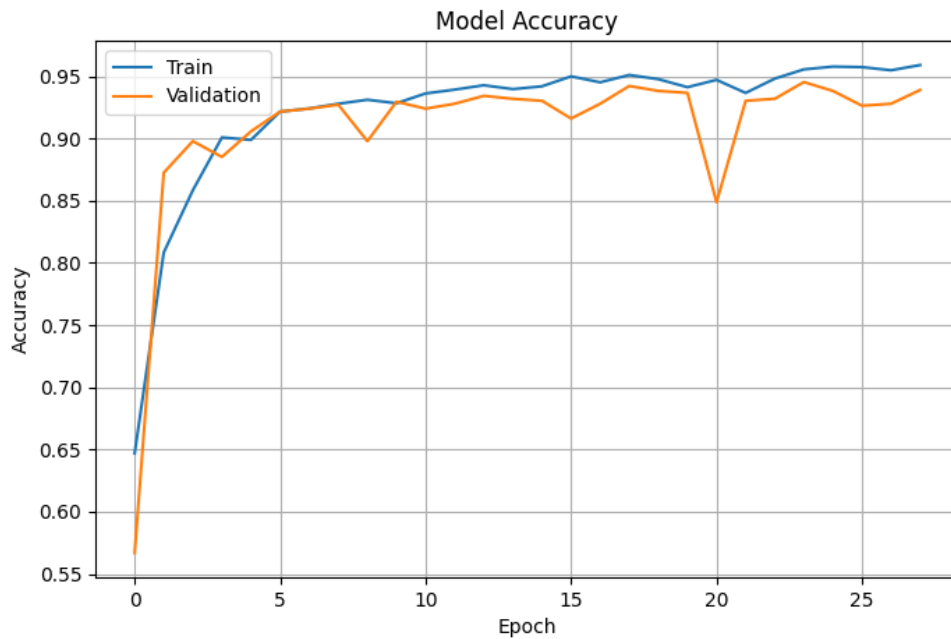


Figure 5.5. Training and validation accuracy of CNN–BiGRU using raw ECG signals.

Figure 5.5 shows the training and validation accuracy of the CNN–BiGRU model using raw ECG signals. The training accuracy increases rapidly during the initial epochs and gradually reaches about 95%, while the validation accuracy remains around 92–94% for most later epochs, with a noticeable temporary drop around the 20th epoch. Overall, the curves indicate that the model learns effectively from the raw ECG signals.

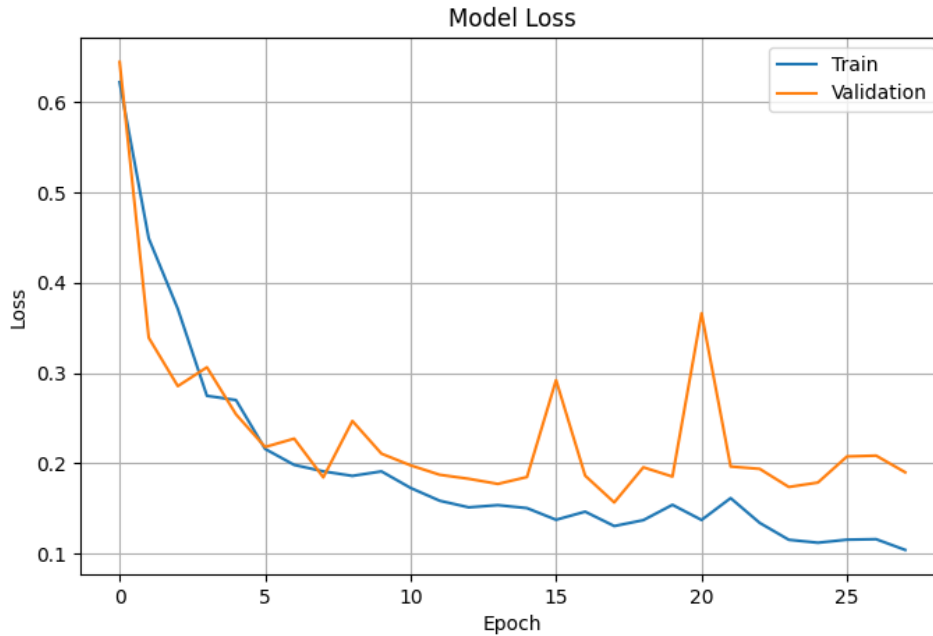


Figure 5.6. Training and validation loss of CNN–BiGRU using raw ECG signals.

Figure 5.6 shows the training and validation loss of the CNN–BiGRU model using raw ECG signals. The training loss decreases progressively throughout the epochs, while the validation loss decreases initially and then fluctuates, including some noticeable increases during later epochs. This behaviour indicates continued improvement in training performance with some variation in validation loss.

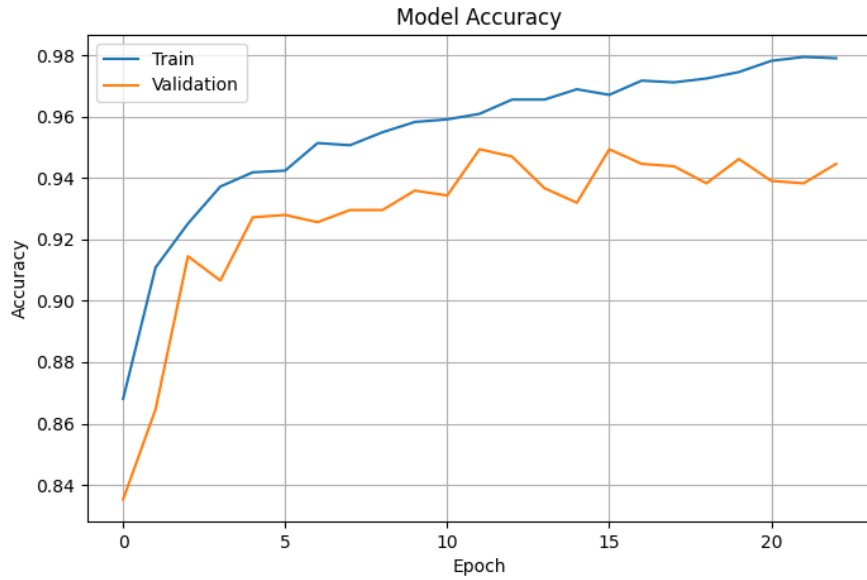


Figure 5.7. Training and validation accuracy of CNN–BiGRU using preprocessed ECG signals.

Figure 5.7 shows the training and validation accuracy of the CNN–BiGRU model using preprocessed ECG signals. The training accuracy increases steadily and reaches approximately 98%, while the validation accuracy remains around 94–95% during the later epochs, with minor fluctuations. This indicates effective learning with relatively stable validation performance.

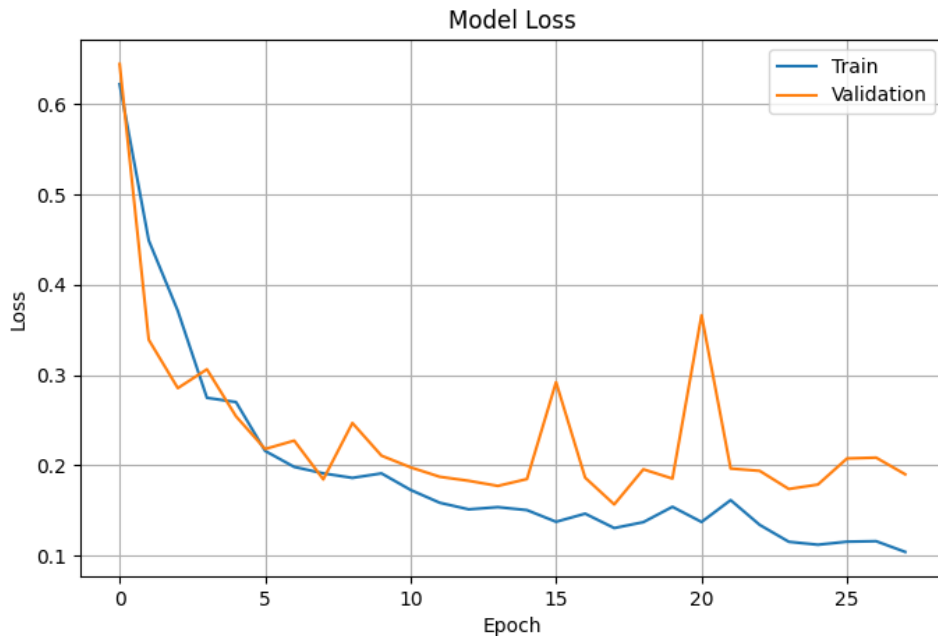


Figure 5.8. Training and validation loss curve for the CNN–BiGRU experiment.

Figure 5.8 shows the training and validation loss of the CNN–BiGRU model using preprocessed ECG signals. The training loss decreases progressively throughout the training process, whereas the validation loss decreases

initially and shows fluctuations during the later epochs. The overall reduction in training loss indicates that the model learns effectively from the preprocessed ECG data.

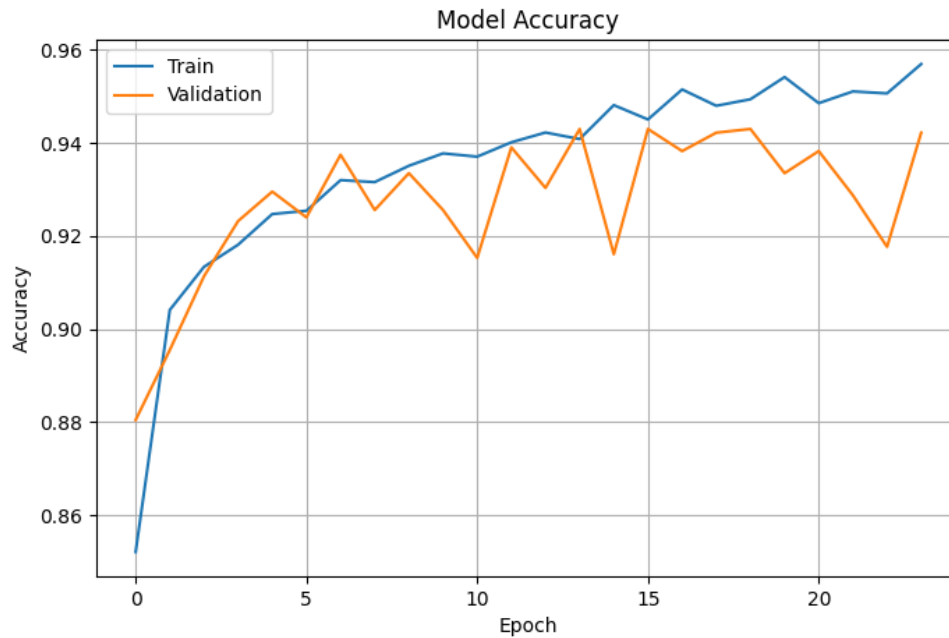


Figure 5.9. Training and validation accuracy of CNN–Positional Encoding–Transformer using raw ECG signals.

Figure 5.9 shows the training and validation accuracy of the CNN–Positional Encoding–Transformer model using raw ECG signals. The training accuracy increases progressively and reaches approximately 95–96%, while the validation accuracy remains around 92–95% with some fluctuations across the epochs. The relatively close training and validation performance indicates stable learning of the model.

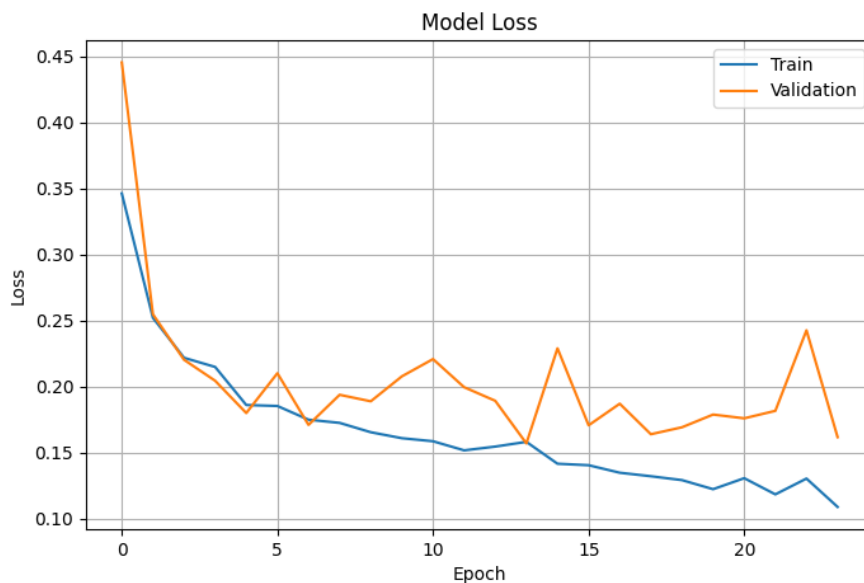


Figure 5.10. Training and validation loss of CNN–Positional Encoding–Transformer using raw ECG signals.

Figure 5.10 shows the training and validation loss of the CNN–Positional Encoding–Transformer model using raw ECG signals. The training loss decreases steadily throughout the epochs, while the validation loss decreases initially and exhibits fluctuations during later epochs. The overall reduction in training loss indicates effective learning, while the variations in validation loss reflect changes in validation performance during training.

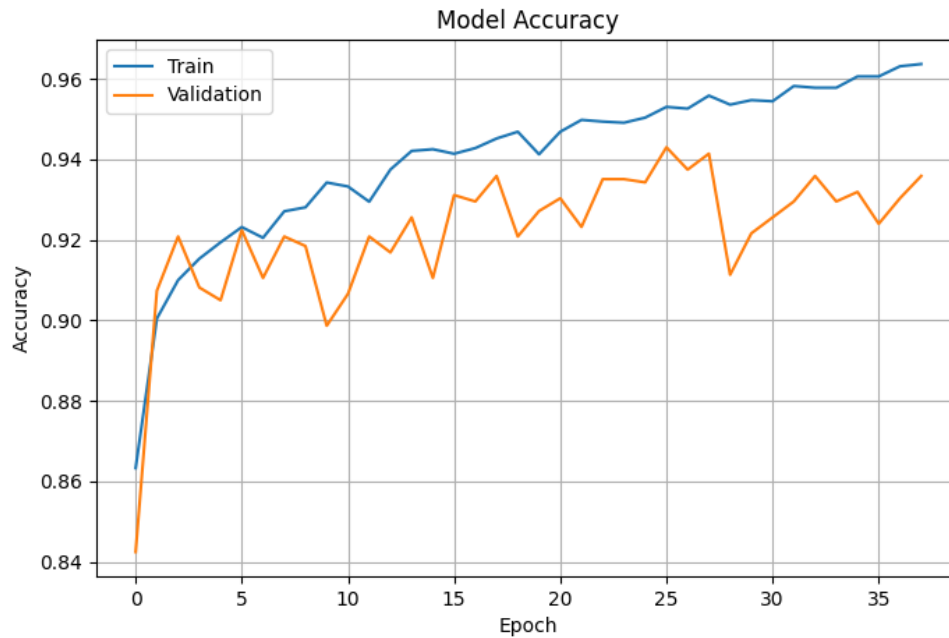


Figure 5.11. Training and validation accuracy of CNN–Positional Encoding–Transformer using preprocessed ECG signals.

Figure 5.11 shows the training and validation accuracy of the CNN–Positional Encoding–Transformer model using preprocessed ECG signals. The training accuracy increases gradually and reaches approximately 96%, while the validation accuracy remains around 92–94% with some fluctuations across the epochs. The overall trend indicates stable model learning with reasonably consistent validation performance.

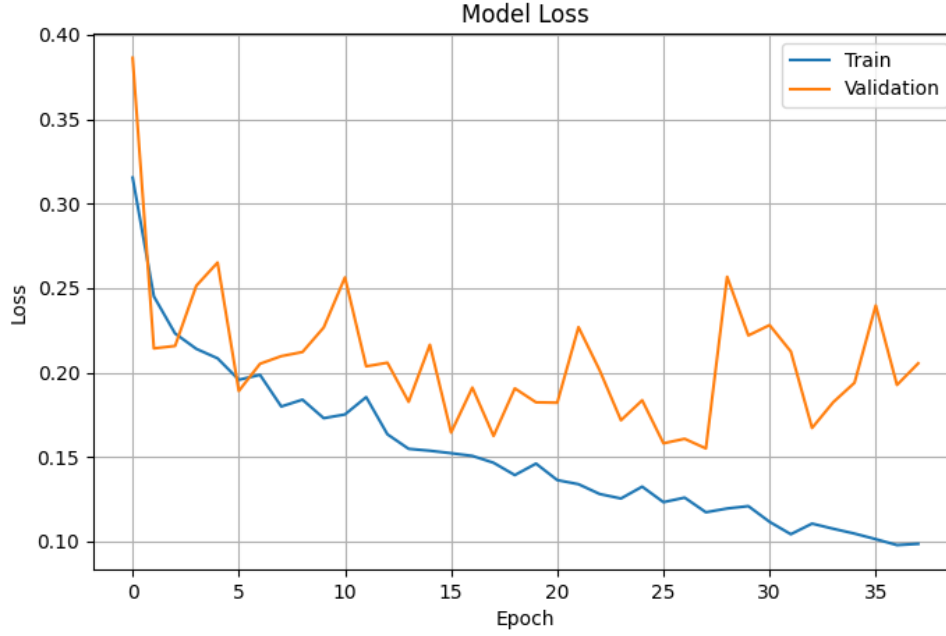


Figure 5.12. Training and validation loss of CNN–Positional Encoding–Transformer using preprocessed ECG signals.

Figure 5.12 shows the training and validation loss of the CNN–Positional Encoding–Transformer model using preprocessed ECG signals. The training loss decreases progressively throughout the training process, whereas the validation loss shows fluctuations while remaining generally lower than its initial value. This indicates effective learning from the preprocessed ECG signals, with some variation in validation performance across the epochs.

## 5.5 Comparative Analysis

The raw and preprocessed experiments show that denoising is not automatically beneficial for downstream MI classification. CNN–BiLSTM achieved a higher reported accuracy on raw ECG (95.54%) than on the primary preprocessed ECG condition (94.25%). The available preprocessed results also show CNN–BiLSTM outperforming CNN–BiGRU in accuracy, recall, F1-score, and ROC-AUC. The Transformer-based model remained competitive, but its performance varied with the preprocessing configuration.

The filtering experiments further indicate that the choice of cut-off frequency can influence validation behaviour and classification sensitivity. In the reported experiments, the 0.5–45 Hz configuration achieved 93.85% accuracy and 92.33% recall, compared with lower recall values reported for some individual filtering configurations. Therefore, preprocessing should be treated as an experimental design variable rather than as a universally beneficial operation.

## 5.6 Discussion of Training Behaviour

Across the training curves, accuracy improves rapidly during the initial epochs and then becomes more stable. In several experiments, training accuracy continues to increase while validation accuracy fluctuates around a narrower range. The corresponding loss curves show a general decline in training loss, with validation loss



exhibiting periodic increases and decreases. This behaviour indicates a degree of generalization gap and suggests that the best-performing epoch may not necessarily correspond to the final training epoch.

## Chapter 6: Conclusion and Future Scope

### 6.1 Summary of Work

This internship implemented and compared three CNN-based hybrid deep learning architectures for binary classification of Normal and MI 12-lead ECG signals. The architectures were designed around a common CNN feature-extraction stage and differed in the sequence-modelling component: BiLSTM, BiGRU, and positional encoding with a Transformer encoder. The study also compared raw ECG input with a primary preprocessing pipeline based on band-pass filtering, 50 Hz notch filtering, and Z-score normalization.

### 6.2 Outcomes Achieved

- A balanced experimental set of 10,000 ECG records was prepared according to the internship configuration.
- Invalid records containing NaN or missing signal values were excluded during data preparation.
- Three hybrid architectures were implemented and evaluated.
- Raw and preprocessed input conditions were compared without changing the corresponding model architecture.
- Additional filtering and cut-off experiments were performed using the Transformer-based architecture.
- Training curves and classification metrics were collected for comparative analysis.

### 6.3 Limitations

- The experiments use a selected subset and binary Normal/MI formulation rather than the full range of cardiovascular diagnoses.
- The current evaluation is based on the selected experimental split and does not establish performance across independent external clinical cohorts.
- Reconstruction quality of preprocessing was not itself the primary target; the main objective was downstream classification.
- Clinical deployment would require additional validation, interpretability analysis, and prospective assessment.

### 6.4 Future Enhancements

- Evaluate the models on independent external ECG datasets to assess generalization.
- Investigate explainability methods for identifying ECG regions or leads contributing to MI predictions.
- Explore lead-wise attention and multi-scale CNN feature extraction.
- Compare additional Transformer and state-space sequence models.
- Perform systematic hyperparameter optimization and class-wise error analysis.
- Evaluate robustness under real-world noise and distribution shifts.

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